
WorldFork: Auditable Branching Socio-Institutional World Models for LLM Forecasting Agents

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Abstract

LLM forecasting systems usually return a probability and rationale, but they rarely expose the counterfactual rollouts, endpoint evidence, social assumptions, and unresolved uncertainty behind that forecast. We present WorldFork, a forecasting protocol that turns a scenario into checkpointed timelines with branch lineage, endpoint ledgers, path mass, report provenance, and explicit socio-institutional state. We build an evaluation package with 108 public initialization and audit cards, a 24-card retrospective resolved forecasting pilot, and long-horizon branch-stress tasks for lineage and report grounding. In the current run, static leakage checks pass, source fetch verification succeeds for 36 of 40 resolution URLs, direct LLM baselines complete on all 24 resolved cards, and default-route WorldFork no-branch rows complete at a 16-tick cap plus a 35-tick resume. The 35-tick resume did not improve frozen path-mass forecasts over the 16-tick rows. The evidence therefore supports WorldFork as an auditable forecast artifact rather than an established accuracy improvement; endpoint-ledger closure and long-horizon branching runs are required before making stronger Brier-score claims.

1. Introduction

Forecasting is increasingly used to test whether general-purpose AI systems can reason under uncertainty about the real world. A useful forecast, however, is more than a scalar probability: proper scoring rules measure accuracy (Brier, 1950; Gneiting & Raftery,

2007), while forecasting practice also depends on exposing assumptions and uncertainty (Tetlock & Gardner, 2015). Institutional and social events depend on actors, authorities, incentives, trust and dependency relationships, procedural deadlines, information flows, and unresolved evidence. Single-shot LLM forecasts can mention these factors in prose, but their internal assumptions are difficult to audit or compare after the fact.

WorldFork addresses this gap by representing a forecast as an auditable branching world model. A scenario is initialized into actors, cohorts, candidate endpoints, graph layers, and initial socio-institutional state. The runtime advances timelines in ticks, forks alternate continuations when mechanisms diverge, tracks endpoint evidence in ledgers, records path mass, and generates reports with provenance. The purpose is not to claim that branching simulation automatically improves probabilistic accuracy. The purpose is to expose the forecast object: which assumptions led to which endpoint mass, where uncertainty remained, and which report claims are grounded in logged artifacts.

This paper makes three contributions. First, it specifies a branching forecast protocol with checkpointed Big Bangs, ticks, branch lineage, endpoint ledgers, path mass, and report provenance. Second, it treats socio-institutional state as audit data: actors, cohorts, trust/dependency/conflict/influence/coalition/exposure graphs, sociology signals, source packets, and bounded emotion observations. Third, it provides a pilot benchmark and evaluation workflow covering initialization quality, resolved forecast scoring, long-horizon audit metrics, and social-state consistency.

2. Method

Initialization. A public scenario card is converted into a Big Bang containing typed actors and cohorts, scenario constraints, candidate endpoints, graph layers, and initial social state. For resolved forecast cards, endpoints are binary yes/no outcomes. For dossier

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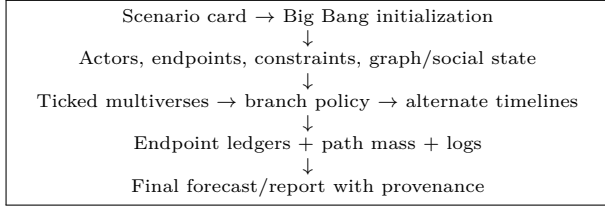


Figure 1. WorldFork represents a forecast as a set of inspectable runtime artifacts rather than only a probability and rationale.

and calibration cards, endpoints include institutional resolutions plus unresolved or abstention states. Initialization is scored for schema completeness, actor recall, authority fidelity, constraint preservation, endpoint extraction, graph quality, prompt-injection resistance, and uncertainty honesty.

Ticks and branches. A multiverse advances in ticks. Each tick updates actor positions, social signals, events, endpoint evidence, and logs. A branch policy controls maximum depth, active multiverses, branches per tick, and branch-score threshold. Short resolved runs compare no-branch and branching policies. Long-horizon runs stress whether lineage, branch locality, path mass, and endpoint ledgers remain inspectable over many ticks.

Endpoint ledgers and reports. Endpoint ledgers track whether candidate endpoints are supported, contradicted, unresolved, or not yet checkable. Reports are scored by sampling claims and verifying support from ticks, ledgers, logs, or source packets. For resolved scoring, yes/no probability is normalized over yes and no when unresolved mass is present; unresolved mass is still reported separately.

Socio-institutional state. WorldFork models social assumptions as audit state, not validated psychology. Graph layers and sociology signals are useful only when they make branch divergence or report claims traceable. Emotion observations are bounded logged variables and should not be interpreted as hidden human affect.

3. Benchmark and Metrics

The benchmark contains 108 public cards: 72 existing synthetic WorldFork stress prompts and 36 added forecasting cards. The add-on contains 24 retrospective resolved forecast cards, 8 longform source-packet dossiers, and 4 adversarial/calibration cards. Private resolutions and gold checklists are withheld from forecast-producing systems until forecasts are frozen.

Forecast accuracy on resolved cards uses binary Brier

Group	Count	Primary use
Synthetic stress cards	72	initialization and audit
Resolved forecast cards	24	Brier/log scoring
Longform dossiers	8	report and ledger grounding
Adversarial calibration	4	uncertainty and injection tests

Table 1. Benchmark composition for the ICML forecasting package.

Condition	n	Brier	Log	Unres.
Direct LLM	24	0.2425	0.7017	0.0000
Structured	24	0.2384	0.6775	0.0000
WF no-branch 16t	24	0.4253	1.6864	0.9236
WF no-branch 35t	24	0.4253	1.6864	0.9236

Table 2. Resolved-card forecast scores. The 35-tick WorldFork row reuses and extends the 16-tick Big Bangs; the extracted path-mass forecasts are unchanged.

score $(p_{\text{yes}} - y)^2$ and clamped negative log score. WorldFork outputs may assign mass to unresolved states; primary yes/no scoring normalizes over yes/no when possible and reports unresolved mass separately. Audit runs use 0–4 rubric scores for lineage integrity, branch locality, endpoint coverage, endpoint honesty, path-mass consistency, report grounding, failure observability, cost transparency, social-state consistency, and emotion observability.

4. Results

Card QA. Static package QA passes. The run contains 108 public cards, 36 private evaluation rows, matching public/private IDs, no private resolution fields in public cards, 24/24 resolved cards with at least one resolution source, and balanced private labels (12 yes, 12 no). Automated source fetching checked 40 resolution URLs: 36 fetched successfully and 4 were blocked or timed out. We therefore treat source validation as mostly complete but not final.

Direct baselines. We ran two retrieval-free OpenAI Codex gpt-5.4 direct baselines on all 24 resolved cards. The unstructured direct prompt achieved mean Brier 0.242492 and mean log score 0.701698. The structured direct prompt achieved mean Brier 0.238363 and mean log score 0.677498. Both baselines have zero unresolved mass because the output contract requires a normalized yes/no distribution.

Live WorldFork evidence. The full 108-card suite now

has live initialization evidence. Initialization on one resolved card completed in 146.20 seconds with 9 actors, 9 traits, graph state, sociology baseline, and emotion baseline present. Queued initializer batches then completed the remaining public cards: the largest add-on batch reached p1's configured concurrency of 8 and drained 22 jobs in 464.64 seconds wall time, while the existing 72-card batch completed in 1796.65 seconds with p1 saturated at 8 for most of the run. The automated coverage table reports 108/108 succeeded initializations, mean actor count 8.76, mean trait count 8.82, mean graph edge count 33.99, and sociology plus emotion baselines present for 108/108 cases; semantic 0-4 initialization quality scoring remains pending.

The default-route no-branch WorldFork run completed all 24 resolved cards at a 16-tick cap, then resumed unresolved Big Bangs to a 35-tick cap using OpenRouter deepseek/deepseek-v4-flash for cohort/hero routes and strong governance/report routes. The 35-tick resume produced final Big Bang reports for 17 frozen rows and endpoint-ledger/path-mass artifacts for all rows, but none of the extracted yes/no/path-mass forecasts changed from the 16-tick output. The unresolved-mass distribution remained high: 21 rows at 1.0, one at 0.6667, one at 0.5, and one at 0.0. This validates the continuation/reuse path and report recovery, but it is negative evidence for the current endpoint-ledger closure behavior.

Paired bootstrap intervals over the 24 resolved card IDs support the same caution. The 35-tick no-branch row has a mean Brier difference of 0.1870 against the structured direct baseline, with a 95% bootstrap interval of [0.0370, 0.3469], and a mean log-score difference of 1.0089 with interval [0.2955, 1.7800]. These intervals are descriptive for the pilot only; branching, long-horizon audit, and social-state rows are still required for the full benchmark.

Queue tuning. The synchronous smoke left Celery queues idle. The queued smoke and scored validations showed active p1 worldfork.execute_job tasks while p0/p2/p3 were idle. The queued initializer batches showed that p1 can run independent cases concurrently up to its configured concurrency of 8, and the three-case E3 no-branch batch confirmed that independent run jobs overlap in one shared wall-clock window. Thus Celery is functioning; queueing improves throughput across cases, while one case remains a long p1 job. The first branching validation shows that branch policy, not queue capacity, is now the main runtime multiplier.

5. Limitations

The resolved cards are retrospective and use partial masking, so leakage cannot be ruled out. The synthetic and dossier cards test auditability and stress behavior rather than direct real-world forecasting accuracy. Social-state metrics are rubric-based and require evidence sampling. Emotion observations are audit variables, not validated human psychology. The default-route no-branch forecast matrix is complete but highly unresolved, and extending the same Big Bangs from 16 to 35 ticks did not change path-mass forecasts. Stronger accuracy or auditability claims require default-route branching rows plus the pending long-horizon and social-state scoring blocks.

6. Conclusion

WorldFork is best viewed as an auditable forecasting substrate. Its value is the production of inspectable forecast artifacts: branch lineage, endpoint ledgers, path mass, report provenance, unresolved uncertainty, and social assumptions that can be checked after the fact. The current evidence supports the feasibility and scoring setup, while the final accuracy and auditability claims should be made only after the full WorldFork conditions are completed.

References

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